

Komal Jha

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Final-year Computer Science student at VIT Vellore who builds, tests and deploys software across the stack, from **real-time web apps** and **distributed API infrastructure** to **machine-learning** systems. Frontend internship at **Tech Mahindra**; six documented projects with live demos.

Education

Vellore Institute of Technology, Vellore — <i>B.Tech in Computer Science (CGPA: 7.51)</i>	2023 – 2027
Swarajya Senior Secondary School, Alwar — <i>Class XII (77.8%)</i>	2023
Saint Francis School B.Deoghar, Deoghar — <i>Class X (95.4%)</i>	2021

Experience

Tech Mahindra, Navi Mumbai	May 2025 – July 2025
<i>Frontend Development Intern (7-week summer internship)</i>	
– Contributed to the frontend of a Corporate Banking Solutions application for a PSU bank. – Developed and debugged UI components using Angular, TypeScript, HTML and CSS. – Resolved styling issues across existing screens to improve UI consistency and responsiveness. – Collaborated with mentors and team members on enhancements aligned with business requirements, following Agile.	

Projects

GateKeeper – Multi-Tenant API Gateway & Rate Limiter Code · Live demo	2026
Tech Stack: Node.js, Express, Redis (Lua), PostgreSQL, React, Recharts	
– Engineered an atomic Redis Lua token-bucket limiter, load-tested race-condition-safe under 20 concurrent requests. – Implemented SHA-256 hashed API keys, instant revocation, per-tenant quotas and a live analytics dashboard.	
SyncBoard – Real-Time Collaborative Kanban Board Code · Live demo	2026
Tech Stack: React, Node.js, Express, MongoDB, Socket.IO, Tailwind CSS	
– Built real-time multi-user sync and live cursors with Socket.IO rooms; deployed on Render and Vercel. – Enforced 3-tier RBAC on both REST and WebSocket layers, closing a live-session access gap.	
StreamVault – Video Upload & Streaming Platform Code	2026
Tech Stack: React, Node.js, Express, MongoDB GridFS, FFmpeg	
– Built a background FFmpeg transcoding pipeline (360p/480p/720p) with range streaming and adaptive quality. – Designed a viewer drop-off analytics dashboard, verified with 16/16 end-to-end tests.	
Dental X-ray Caries Classifier Code · Live demo	2026
Tech Stack: Python, PyTorch, torchvision, OpenCV, scikit-learn, Streamlit	
– Fine-tuned ResNet-18 via transfer learning on 5,000 X-rays; reached ROC-AUC 0.74 on 750 unseen images. – Implemented Grad-CAM from scratch to explain predictions; reported bootstrap 95% CIs and honest limitations. – Deployed a live Streamlit demo returning class probabilities and a Grad-CAM overlay per upload.	
Heart Disease Prediction Pipeline Code · Live demo	2026
Tech Stack: Python, scikit-learn, XGBoost, Pandas, NumPy, Altair, Streamlit	
– Built a leak-free pipeline and compared 5 models under identical repeated stratified CV; tuned the top 2 with nested CV. – Reached ROC-AUC 0.92 ± 0.04 and 85.8% accuracy; explained predictions with permutation importance. – Deployed a live, interactive Streamlit app with a risk gauge and results dashboard.	
CareerVault – Job Application Tracker Code	2026
Tech Stack: React, Node.js, Express, MongoDB Atlas, JWT, Recharts	
– Built a MERN tracker with JWT auth, bcrypt hashing, REST APIs and a Recharts dashboard with search, filters and pagination.	

Skills

- **Languages:** Python, Java, JavaScript, SQL, C++, C
- **Frontend & Backend:** React, Angular, Node.js, Express, REST APIs, WebSockets, JWT, RBAC
- **Data & ML:** MongoDB, PostgreSQL, Redis, PyTorch, scikit-learn, XGBoost, Pandas, Grad-CAM
- **Tools & Platforms:** Git, Postman, Streamlit, Vercel, Render, MongoDB Atlas
- **Core Concepts:** Data Structures & Algorithms, OOP, System Design, Concurrency, Agile

Certifications

AWS Certified Cloud Practitioner — Amazon Web Services	2026
Data Structures & Algorithms — Udemy	2026